ANSWER KEY

Quiz: Lesson 2 — Data Visualization and Operational Dashboards

Course: Productivity Tools for Technical Reporting

Questions

Multiple Choice

****Q1.** You are building the OrbitCo 30-day review dashboard and need a visual that shows how weekly ticket volume has evolved over the month. Which chart type is the most appropriate?**

- [ ] a) Pie chart with one slice per week
- [ ] b) Scatter plot of week number vs ticket count
- [x] **c) Line chart with week on the x-axis and ticket count on the y-axis**
- [ ] d) Stacked column chart with weeks stacked inside a single bar

Correct: c

> A line chart is the canonical visual for a continuous metric over ordered time, which is exactly what weekly ticket counts over a month look like. Option (a) misuses a pie — pies show composition of a whole, not trends. Option (b) treats week number as a continuous variable but loses the connectedness that makes the trend readable. Option (d) is structurally wrong — stacking weeks inside a single bar hides the time axis. "Name the question" discipline: the question is "how did volume change week-to-week," and line charts answer exactly that.

Q2. On the OrbitCo dashboard, the Uptime % Key Performance Indicator (KPI) tile shows the value `99.92%` and carries an icon-set traffic-light rule with thresholds `≥ 99.9 → green`, `≥ 99.5 → yellow`, `otherwise → red`. The icon is rendering red on every reload. What is the most likely cause?

- [ ] a) The cell value is stored as text, not a number
- [x] **b) The thresholds are set to the percentage form (99.9, 99.5) but the underlying cell value is a decimal (0.9992), so the `≥` comparisons fail**
- [ ] c) Icon sets do not support three thresholds; only two are allowed
- [ ] d) Conditional formatting does not work on cells with percentage formatting

Correct: b

> The cell displays `99.92%` because it is formatted as a percentage, but the underlying value is `0.9992`. Comparisons in the icon-set rule use the underlying value, not the display. Thresholds must be in decimal form (`0.999`, `0.995`) for the condition to fire correctly. Option (a) would produce a different failure mode (no icon at all). Option (c) is false. Option (d) is false. This is the single most common dashboard bug and worth catching the first time.

Q3. Which of the following is the most important principle of operational dashboard layout?

- [ ] a) Pack as many charts as possible onto the dashboard so readers have every view available
- [ ] b) Keep the raw `Tickets` Table visible on the dashboard sheet so readers can verify numbers directly
- [x] **c) Use the three-band pattern — KPI tiles on top, trends in the middle, drill-down details on the bottom — with raw data on separate sheets**
- [ ] d) Use a different color palette on every chart so readers can visually distinguish them

Correct: c

> The three-band layout matches how readers actually scan a page (headlines first, trends second, evidence third) and the separation of raw data from the dashboard sheet keeps the one-page view uncluttered. Option (a) produces visual noise; more charts usually means less insight. Option (b) violates the dashboard-hygiene rule about raw data and makes the dashboard feel like an analysis workspace rather than a summary. Option (d) confuses the reader — consistent palettes make dashboards easier to read, not harder.

Multiple Select

****Q4.** Which of the following are specific misleading-visualization patterns that an operations analyst should recognize and avoid in dashboard reporting? * (Select all that apply.)***

- [x] **a) Truncated y-axis on a bar or column chart**
- [x] **b) 3D perspective applied to bar, column, or pie charts**
- [ ] c) Using a horizontal bar chart for a categorical breakdown with long labels
- [x] **d) Using a line chart to show a trend across unordered categorical values**
- [ ] e) Sorting horizontal bars by category value to make magnitude comparison easy
- [x] **f) Using red and green for non-ordinal categorical data with no pass/fail meaning**

Correct: a, b, d, f

> The four misleading patterns from the lesson are (a) truncated y-axis exaggerating small differences, (b) 3D distortion biasing slice/bar sizes, (d) line charts implying connection between unordered categories where none exists, and (f) red/green on non-ordinal categories implying a pass/fail meaning that is not present. Options (c) and (e) are both legitimate design choices — horizontal bars handle long labels well, and sorting by value improves readability. Being able to distinguish honest design decisions from misleading ones is half of the critical-eye skill this lesson builds.

True/False

Q5. When pasting a chart from Excel into a PowerPoint deck that you will hand off to a colleague who does not have access to the source Excel file, **Paste as Picture** is the correct choice.

Correct: True

> Paste as Picture produces a static image that preserves the exact rendering and is fully portable — it does not depend on the source Excel file being reachable. Paste with Link would break the moment the colleague opened the deck because they do not have access to the source. Paste as Embedded Object would bloat the file and risk format drift as the embedded chart's styles diverge from the original. The practical rule from the lesson is "if you will not be the one opening the deck when it is used, paste as picture," and this scenario is the canonical case for that rule.